A Comparative Study of Risk and Return Using Markowitz and Single Index Models: Evidence from Select NIFTY 50 Stocks

Subject: Securities Analysis and Portfolio Management

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1. Introduction

1.1. Objective of the Report

The goal of this report is to assess and evaluate the investment quality of five companies selected from the NIFTY 50 index. Each company represents a separate sector. We will arrive at an analysis of the companies using company-level, fundamental analysis and evaluate the risks/returns of the stocks by modelling them as a portfolio of assets.

The report focuses on:

- Evaluate the companies' finances and performance trends through key financial ratios using S. Kevin's SAPM framework over the previous five years.
- Determine the expected returns and risks of each stock by determining their daily price movements for the previous year. This will be based on statistical analysis and excel calculations.
- Estimate the compound, diversified portfolio of the 5 stocks, equally weighted and evaluate the portfolio expected maximum return and risk:
    - Using Markowitz's Modern Portfolio Theory (MPT), which accounts for the covariance of the assets.
    - Using the Single Index Model (SIM), specifically the regression-based estimation of alpha and beta.
- Compare the results of both the modelling performance expectations in order to determine which model produces an efficient risk-return trade-off, based on assumptions.
- Provide investing insights based on our financial and quantitative analysis with respect to improving investment decisions and risk management for a diversified equity investor.

1.2. Methodology and Data Sources

A. Methodology:

This method follows a structured methodology which builds on fundamental analysis and quantitative portfolio models to examine the investment performance of the five NIFTY 50 stocks.

- Company Analysis (Fundamental Analysis):
    - The report focused on assessing each of the selected companies financial health using selected financial ratios as outlined in the Security Analysis and Portfolio Management textbook by S. Kevin.
    - Ratios assessed included profitability, liquidity, leverage, valuation and efficiency ratios calculated from the last five years financial data.
- Return and Risk Estimation:
    - Daily close price data over the last one year was used to calculate:
        - Daily returns
        - Expected Annual Return (Mean Daily Return * 252)

        - Annualized Risk (Standard Deviation of Daily Returns * √252)
- Portfolio Analysis with the Markowitz Model:
    - An equally weighted portfolio of 5 individual stocks was constructed.
    - The portfolio risk and return were calculated using:
        - Covariance Matrix
        - Portfolio Variance and Standard Deviation
        - Expected Portfolio Return (Individual Weighted Average Returns)
- Portfolio Analysis using Single Index Model (SIM):
    - Performed regression analysis of each stock's return against NIFTY 50 Index to determine:
        - Alpha (α), Beta (β), Residual risk (ε)
    - Calculate expected returns using the CAPM based formula:
        - $E(R_i)=R_f+\beta_i(R_m-R_f)$
    - Evaluated portfolio return and risk based on model-based beta and residual variance.
- Comparison and Interpretation:
    - Compared portfolio risk-return outcomes using both the models.
    - Interpreted the practical relevance and efficiency of each model in constructing diversified portfolios.

B. Data Sources:

- Company Financial Statements:
    - Annual Reports (2019-2024) from the company websites, NSE India and Moneycontrol.
- Stock Price Data:
    - NSE India
    - Yahoo Finance
    - Moneycontrol
- Index Data (Benchmark):
    - NIFTY 50 Daily Returns (Used as Market Index)
- Risk Free Rate:
    - 1-year T-Bill rate disclosed in the RBI website (Used for CAPM/SIM)
- Text book materials:
    - Security Analysis and Portfolio Management by S. Kevin for definitions of ratios and theoretical models.

1.3. Timeframes

For the purpose of this analysis, the following timelines have been adopted for the project purposes and model requirements.

- Fundamental (Company) Analysis:
    - Time Period: FY 2020-21 to FY 2024-25 (Five Financial Years)
    - Purpose: To evaluated longer-term financial performances and trends by employing ratio analysis.

- Source Data: Audited Financial Statements (P&L, Balance Sheet and Cash Flow).
- Return and Risk Analysis:
  - Time Period: 1 Year of Daily Closing Prices, from 26th July, 2024 to July 25th, 2025.
  - Purpose:
    - To calculate daily returns, annualized expected return and risk (standard deviation) for each stock.
    - Required in order to compute Markowitz Model and Single Index Model.
- Index and Risk-Free Rate:
  - Indicator Index: NIFTY 50 (To be used in the SIM regression for market return)
  - Risk-Free Rate: 1 year Treasury Bill Rate (As of 30 June 2025) obtained from the RBI website.

1.4. List of 5 selected NIFTY stocks

Company Name	Sector
NTPC Limited	Power / Utilities
Dabur	FMCG / Consumer Goods
HDFC Life Insurance	Insurance / BFSI
Adani Enterprises	Infrastructure / Conglomerate
Hindalco Enterprises	Metals / Mining

Table 1: List of NIFTY 50 Stocks

These companies were selected to ensure sectoral diversification for the portfolio and to explore varied financial dynamics across industries.

2. Company Analysis (Fundamental Analysis)

2.1. Brief overview of each company

- **NTPC Limited (Power and Utilities)**

  NTPC is India's largest energy conglomerate and Maharatna Central Public Sector Undertaking which was established in 1975. With a presence across power generation such as coal, hydro, nuclear and renewables, it contributes over 25% of the India's total power output and holds around 17% of the national capacity as on 2023 March (*NTPC Overview | NTPC Limited*, n.d.). It comprises of diversified operations which comprises of consultancy, power trading, coal mining, rural electrification, and ash utilization. (*NTPC Overview | NTPC Limited*, n.d.)

- **Dabur (FMCG-Consumer Goods)**

  Dabur was started in 1884, and is one of India's oldest FMCG companies based on Ayurvedic and natural products. The company has healthcare, personal care, home care, and foods & beverages segments operating a wide range of brands including Chyawanprash, Real, Vatika, and Dabur Red Paste(*World's Best Ayurvedic Brand: Plastic Waste Positive FMCG Company | Dabur*, n.d.). Dabur is present in 120+ countries and conducts business with 7 million retail outlets and generates about 25% of its revenue from markets outside of India. In India, the company operates 13 manufacturing plants in the country with 8 facilities globally. For fiscal year 2024–25, the company reported revenue of ₹12,563 crore and a net profit of ₹1,767 crore with a flat rural growth rate due to continued margin pressures (*Dabur India Ltd Company Profile - Dabur India Ltd Overview - GlobalData*, n.d.). The company is in a process of strategic restructuring in partnership with McKinsey. In this engagement, Dabur is looking to develop premiumization, expansion of digital and pruning of their portfolio by exiting low margin categories, for example, tea and diapers (*Dabur India Ltd Company Profile - Dabur India Ltd Overview - GlobalData*, n.d.).

  Notable acquisitions include:
  - Badshah Masala (2022) – Spices & condiments
  - Sesa Care (2024) – Personal care & cosmetics

  Dabur has announced it will achieve carbon neutrality by 2040 and is plastic waste positive already.

- **HDFC Life Insurance (Insurance/BFSI)**

  HDFC Life was incorporated in 2000 and is headquartered in Mumbai. HDFC Life is a joint venture between HDFC Ltd and Abrdn (formerly known as Standard Life) and is one of the largest life insurers in India. HDFC Life underwrites a variety of long-term insurance products in India that include term insurance, ULIPs, annuities, pension plans, and endowment plans ("HDFC Life – One of the Best Life Insurance Companies in India," n.d.). HDFC Life products can be sold with a variety of distribution channels

including bancassurance, agents, brokers, and digital channels in more than 980 cities and villages("HDFC Life – One of the Best Life Insurance Companies in India," n.d.).

- **Adani Enterprises Limited (Infrastructure/Conglomerate)**

  Founded in 1993 as Adani Exports, as the flagship of Adani Group, the company serves as the "incubator" for emerging businesses. Today it is involved in coal and iron ore mining, airport operations, roads, water treatment, data centres, solar manufacture, defence & aerospace, edible oil (Adani Wilmar), and resource management, among others (*Businesses | Adani Enterprises Ltd*, n.d.). While the coal and traditional energies sectors remain large, emerging growth drivers include airports and green energy, supported by new institutional capital(*Adani Enterprises Profit Rockets on Growing Airports and Green Business*, n.d.).

- **Hindalco Industries Limited (Metals and Mining)**

  Hindalco, the metals flagship of Aditya Birla Group, is amongst the largest aluminium producers in Asia (outside China) and a leading global producer of copper and aluminium value-added products (*Hindalco Industries Ltd: Leading Aluminium & Copper Producer*, n.d.). The company's operations cover bauxite mining, alumina refining, coal mining, smelting, and downstream rolling products (via Novelis). Hindalco operates 52 units in 10 countries and serves diverse sectors including aerospace, renewables, packaging, and EVs (*Hindalco Industries Ltd: Leading Aluminium & Copper Producer*, n.d.). The company is investing ₹45,000 crore in the coming years to ramp-up its aluminium and copper capacities, in particular in EV and speciality alumina applications(**Hindalco's ₹45,000 Crore Investment Plan* : R/FuturesFundamentals*, n.d.).

2.2. 5-year financial highlights

This section presents the five-year financial highlights for NTPC Limited, Dabur India Ltd., HDFC Life Insurance, Adani Enterprises, and Hindalco Industries, based on their consolidated financial statements from FY 2020-21 to FY 2024-25. The metrics include Sales Turnover (or Total Income for banking and insurance firms), Net Profit, Total Assets, Shareholders' Equity, Borrowings, and Earnings Per Share (EPS), reported in Rs. Cr unless stated otherwise. For each metric, the 2021 and 2025 values are shown with the percentage change, followed by an interpretation of the impact and significance. A tabular summary provides a qualitative overview of trends without numerical values.

a. NTPC Limited

- **Sales Turnover:** 2021: 111531.15 Rs. Cr, 2025: 188138.06 Rs. Cr, 68.7% increase. Interpretation: The significant revenue growth strengthens NTPC's financial capacity, likely enabling increased investments or operational stability, reflecting strong demand or expansion.
- **Net Profit:** 2021: 14578.40 Rs. Cr, 2025: 19720.54 Rs. Cr (Net P/L After Minority Interest & Share Of Associates), 35.3% increase.

Interpretation: The notable earnings growth enhances profitability, supporting potential dividends or reinvestment, indicating improved margins or efficiency.

- **Total Assets:** 2021: 398966.21 Rs. Cr, 2025: 524164.59 Rs. Cr, 31.4% increase. *Interpretation:* The substantial asset growth reflects expanded operations, likely in infrastructure, boosting NTPC's market presence and long-term capacity.
- **Shareholders' Equity:** 2021: 125738.47 Rs. Cr, 2025: 184071.16 Rs. Cr (Total Shareholders' Funds), 46.4% increase.
  Interpretation: The stronger equity base improves financial stability, reducing debt reliance and enhancing investor confidence.
- **Borrowings:** 2021: 197236.53 Rs. Cr (181271.91 Long Term + 15964.62 Short Term), 2025: 247575.12 Rs. Cr (201053.88 Long Term + 46521.24 Short Term), 25.5% increase.
  Interpretation: The moderate debt increase, slower than assets, suggests balanced borrowing to support growth, maintaining financial discipline.
- EPS: 2021: 14.39 Rs., 2025: 20.34 Rs.(calculated as 19720.54 / 96966.7 lakhs, 41.3% increase.
  Interpretation: The improved EPS boosts shareholder value, signalling higher per-share profitability and potential for better returns.

b. Dabur India Limited

- **Sales Turnover:** 2021: 9,561.65 Cr, 2025: 12,563.09 Cr, 31.4% increase.
  Interpretation: The steady rise in revenue indicates consistent market demand and business growth. This suggests effective sales strategies and product reach over the years.
- **Net Profit:** 2021: 1,694.95 Cr, 2025: 1,740.42 Cr, 2.7% increase
  Interpretation: Despite significant growth in revenue, net profit has remained largely flat, indicating rising costs or margin pressures. This calls for better cost efficiency or margin control strategies.
- **Total Assets:** 2021: 10,847.13 Cr, 2025: 16,232.28 Cr, 49.6% increase.
  Interpretation: The strong increase in assets reflects expansion in infrastructure, investments, and operational capacity, positioning Dabur for long-term growth.
- **Shareholders' Equity:** 2021: 7,663.53 Cr, 2025: 10,800.69 Cr (Total Shareholders' Funds), 40.9% increase.
  Interpretation: The growing equity base strengthens financial stability, indicating healthy internal funding and retained earnings for reinvestment.
- **Borrowings:** 2021: 483.27 Cr (134.13 + 349.14), 2025: 730.12 Cr (304.03 + 426.09), 51.1% increase.
  Interpretation: Although borrowings have increased significantly, they are still low relative to equity and assets. The company maintains a conservative debt profile, which is favourable from a credit and risk standpoint.
- **EPS:** 2021: 9.59, 2025: 9.97, 4.0% increase.
  Interpretation: The modest EPS growth aligns with the flat profit trend. While EPS is stable, it may underwhelm investors seeking stronger earnings momentum.

c. **HDFC Life Insurance**

- **Total Income:** 2021: 71820.15 Rs. Cr, 2025: 97338.93 Rs. Cr, 35.5% increase. *Interpretation*: The significant revenue growth strengthens the premium base, likely from higher policy sales, supporting long-term liability coverage.
- **Net Profit:** 2021: 1468.42 Rs. Cr, 2025: 1759.5 Rs. Cr (Net P/L After Minority Interest & Share Of Associates), 19.8% increase.
  Interpretation: The modest earnings growth improves profitability, enabling better reserves or policyholder returns, though slower than revenue growth.
- **Total Assets:** 2021: 179641.26 Rs. Cr, 2025: 349411.58 Rs. Cr, 94.5% increase. *Interpretation:* The near-doubled assets reflect significant expansion in investments or policies, enhancing market and operational scale.
- **Shareholders' Funds:** 2021: 11192.83 Rs. Cr, 2025: 16153.98 Rs. Cr (Total Shareholders' Funds), 44.3% increase.
  Interpretation: The stronger equity base bolsters financial stability, supporting growth without heavy debt reliance, improving investor trust.
- **Borrowings:** 2021: 600 Rs. Cr, 2025: 2950 Rs. Cr (Long Term Borrowings), 391.7% increase.
  Interpretation: The significant percentage increase in debt, though low in absolute terms, suggests strategic borrowing for expansion with minimal leverage impact.
- **EPS:** 2021: 6.73 Rs., 2025: 8.41 Rs., 25.0% increase.
  Interpretation: The modest EPS growth improves shareholder value, indicating stable per-share profitability despite slower growth than revenue.

d. **Adani Enterprises**

- **Sales Turnover:** 2021: 39537.13 Rs. Cr, 2025: 97894.75 Rs. Cr, 147.6% increase. *Interpretation:* The rapid revenue growth reflects aggressive business scaling, likely across diverse operations, significantly enhancing financial capacity.
- **Net Profit:** 2021: 1304.65 Rs. Cr, 2025: 4072.22 Rs. Cr (Net P/L After Minority Interest & Share Of Associates), 212.2% increase.
  Interpretation: The tripling of earnings strengthens profitability, enabling reinvestment or debt servicing, driven by successful expansion.
- **Total Assets:** 2021: 51642.86 Rs. Cr, 2025: 198135.54 Rs. Cr, 283.7% increase.
  Interpretation: The nearly quadrupled assets indicate massive business growth, likely in infrastructure or new ventures, boosting operational scope.
- **Shareholders' Funds:** 2021: 17158.57 Rs. Cr, 2025: 47689.97 Rs. Cr (Total Shareholders' Funds), 177.9% increase.
  Interpretation: The significant equity growth enhances financial resilience, supporting expansion and reducing relative debt dependence.
- **Borrowings:** 2021: 15293.31 Rs. Cr (9523.3 Long Term + 5770.01 Short Term), 2025: 76236.41 Rs. Cr (64601.26 Long Term + 11635.15 Short Term), 398.5% increase.

Interpretation: The sharp debt increase reflects heavy borrowing to fund growth, potentially raising leverage risks but aligned with asset expansion.

- **EPS:** 2021: 5.67 Rs., 2025: 57.11 Rs., 907.2% increase.
  Interpretation: The dramatic EPS growth significantly enhances shareholder value, reflecting exceptional per-share profitability from expansion**.**

e. Hindalco Inudstries

- **Sales Turnover:** 2021: 131985 Rs. Cr, 2025: 238496 Rs. Cr, 80.7% increase. *Interpretation:* The strong revenue growth bolsters financial capacity, likely from higher production or prices, supporting operational and investment needs.
- **Net Profit:** 2021: 5674 Rs. Cr, 2025: 16880 Rs. Cr (Net P/L After Minority Interest & Share Of Associates), 197.5% increase.
  Interpretation: The near-tripling of earnings enhances profitability, providing resources for growth or shareholder returns, driven by operational efficiency.
- **Total Assets:** 2021: 189699 Rs. Cr, 2025: 265991 Rs. Cr, 40.2% increase. *Interpretation:* The significant asset growth reflects expanded operations, likely in manufacturing, strengthening market position and capacity.
- **Shareholders' Funds:** 2021: 66529 Rs. Cr, 2025: 123709 Rs. Cr (Total Shareholders' Funds), 85.9% increase.
  Interpretation: The robust equity growth improves financial stability, reducing debt reliance and enhancing investor confidence.
- **Borrowings:** 2021: 65014 Rs. Cr (58985 Long Term + 6029 Short Term), 2025: 61931 Rs. Cr (56217 Long Term + 5714 Short Term), 4.8% decrease.
  Interpretation: The slight debt reduction suggests improved financial discipline, lowering interest burdens and enhancing balance sheet strength.
- **EPS:** 2021: 15.63 Rs., 2025: 72.07 Rs., 361.2% increase.
  Interpretation: The substantial EPS growth significantly boosts shareholder value, reflecting strong per-share profitability from operational success.

2.3. Ratio Analysis

This section provides a detailed interpretation of 31 financial ratios for each company from 2021 to 2025, categorized into liquidity, leverage, profitability, activity, and risk assessment ratios. Each ratio is followed by an interpretation of its significance and trend.

A. NTPC Limited

NTPC Ratio Analysis						
Ratio Type	**Ratio Formula**	**2025**	**2024**	**2023**	**2022**	**2021**
Liquidity Ratios						
Current Ratio	= Current Assets/Current Liabilities	1.069663815	0.9575416	0.9277923	0.9052014	0.9322893
Quick Ratio	= (Current Assets - Inventory - Prepaid Expenses)/Current Liabilities	0.890507649	0.7814965	0.7650182	0.7766455	0.7985272
Leverage Ratios						
Debt-Equity Ratio	= Long Term Debt/Shareholders Equity	1.09226171	1.1835968	1.2779181	1.3434747	1.4416583
Debt to Total Assets Ratio	= Total Debt/Total Assets	0.472323245	0.4894668	0.492813	0.5031137	0.494369
Proprietary Ratio	= Shareholders Equity/Total Assets	0.351170536	0.3346739	0.3296325	0.3250139	0.3151607
Interest Coverage Ratio	= EBIT/Interest	4.110516575	4.2406889	4.2782945	4.3237061	3.6831434
Profitability Ratios related to Sales						
Gross Profit Ratio	= Gross Profit (Sales-COGS) / Sales	43.11%	41.72%	39.84%	43.20%	33.25%
Operating Profit Ratio	= EBIT/Sales	28.77%	28.62%	27.09%	30.36%	30.46%
Net Profit Ratio	= EAT/Sales	11.56%	10.74%	9.16%	11.80%	12.51%
Administrative Expenses Ratio	= Admin Expenses/Sales	0.000%	0.048%	0.027%	0.000%	0.009%
Selling Expenses Ratio	= Selling Expenses/Sales	0.000%	0.048%	0.027%	0.000%	0.009%
Operating Expenses Ratio	= (Admin Expenses + Selling Expenses) / Sales	0.000%	0.048%	0.027%	0.000%	0.009%
Operating Ratio	= (COGS+Operating Expenses) / Sales	56.89%	58.33%	60.19%	56.80%	66.76%
Profitability Ratios related to Investment						
Return on Assets	= EAT/Total Assets	4.147%	3.993%	3.617%	3.759%	3.497%
Return on Capital Employed	= EBIT/Total Capital Employed	5.180%	5.075%	4.500%	4.637%	4.284%
Return on Equity	= EAT/Shareholders Equity	11.810%	11.932%	10.973%	11.565%	11.095%

NTPC Ratio Analysis						
Ratio Type	**Ratio Formula**	**2025**	**2024**	**2023**	**2022**	**2021**
Profitability Ratios related to Equity Shares						
Earnings Per Share	= Net Profit Available to Equity Shareholders/Number of Equity Shares	22.42	19.78	16.64	16.15	14.39
Earnings Yield	= EPS/Market Price Per Share	6.27%	5.56%	9.50%	11.96%	13.50%
Dividend Yield	= DPS/MPS	0.00%	2.11%	2.43%	2.96%	5.35%
Dividend Payout Ratio	= DPS/EPS	0.00%	0.38%	0.26%	0.25%	0.40%
Price to Earnings Ratio	= MPS/EPS	15.95	17.99	10.52	8.36	7.41
Profitability Ratios related to Earning Power						
Return on Investment (ROI)	= EAT/Total Assets	4.147%	3.993%	3.617%	3.759%	3.497%
Activity or Efficiency Ratios						
Current Assets Turnover	= Sales/Current Assets	1.683072574	1.8212642	2.1708828	1.858273	1.6312752
Fixed Assets Turnover	= Sales/Fixed Assets	0.50534556	0.5150083	0.5346029	0.419774	0.3708419
Total Assets Turnover	= Sales/Total Assets	0.358929358	0.3717246	0.3950638	0.3185209	0.2795504
Inventory Turnover	= Sales / Average Inventory	10.24120814	11.066566	14.455241	13.300918	10.859126
Debtors Turnover	= Sales / Average Debtors	5.527783613	5.7418952	6.27436	5.8885414	6.6812765
Assessment of Risk						
DTL	= Contribution/EBT	2.66	2.51	2.56	2.44	2.61
DOL	= Contribution/EBIT	1.474795377	1.2959007	1.2350844	1.2666854	1.2722078
DFL	= EBIT/EBT	1.805177788	1.9365938	2.0703847	1.9286451	2.0490492

Figure 1: *Ratio Analysis of NTPC Limited from Excel Workings*

- **Current Ratio**

  Interpretation: The current ratio improved from 0.93 in 2021 to 1.07 in 2025, indicating an enhanced ability to meet short-term obligations. However, the ratio remains marginally above 1, suggesting tight liquidity and limited short-term cushion.

- **Quick Ratio**

  Interpretation: The quick ratio rose from 0.80 in 2021 to 0.89 in 2025. This consistent growth shows improved liquid asset management, although the ratio still reflects modest liquidity without relying on inventory or prepaid expenses.

- **Debt-Equity Ratio**

  Interpretation: The ratio declined from 1.44 in 2021 to 1.09 in 2025, indicating a strategic reduction in financial leverage and greater reliance on equity funding over time. This enhances the firm's financial stability and creditworthiness.

- **Debt to Total Assets Ratio**

  Interpretation: A gradual decline from 0.49 in 2021 to 0.47 in 2025 shows prudent debt management. The company is slightly decreasing its dependence on external debt, contributing to long-term financial sustainability.

- **Proprietary Ratio**

  Interpretation: This ratio increased from 0.32 in 2021 to 0.35 in 2025, demonstrating a gradual strengthening of equity in the capital structure. A higher proprietary ratio indicates greater shareholder confidence and financial resilience.

- **Interest Coverage Ratio**

  Interpretation: Though stable, the ratio decreased from 4.32 in 2022 to 4.11 in 2025. Despite the slight decline, the firm still comfortably meets its interest obligations, suggesting manageable financial risk.

- **Gross Profit Ratio**

  Interpretation: The gross profit ratio has shown a consistent upward trend from 33.25% in 2021 to 43.11% in 2025. This implies improved cost efficiency in managing direct expenses and enhanced pricing power**.**

- **Operating Profit Ratio**

  Interpretation: The ratio remained stable around 28–30% across five years, indicating a consistent core operational profitability irrespective of minor revenue or cost fluctuations**.**

- **Net Profit Ratio**

  Interpretation: An increase from 12.51% in 2021 to 11.56% in 2025 was observed with some interim fluctuations. This reflects an overall steady bottom-line efficiency and control over indirect expenses and taxes.

- **Administrative Expenses Ratio**

  Interpretation: This remained negligible across the five years, indicating highly efficient administrative cost control.

- **Selling Expenses Ratio**

  Interpretation: Similar to administrative expenses, selling expenses were insignificant throughout, aligning with the company's nature as a utility provider with low marketing expenditure.

- **Operating Expenses Ratio**

  Interpretation: This remained consistently low and negligible, reinforcing NTPC's operational efficiency and minimal overhead burden.

- **Operating Ratio**

  Interpretation: The ratio improved from 66.76% in 2021 to 56.89% in 2025, suggesting more efficient cost management and better operating margins over time.

- **Return on Assets (ROA)**

  Interpretation: ROA improved modestly from 3.50% in 2021 to 4.15% in 2025, showing incremental enhancement in asset utilization to generate earnings.

- **Return on Capital Employed (ROCE)**

  Interpretation: ROCE increased from 4.28% to 5.18% over the period, reflecting a positive trend in the company's ability to generate returns from its overall capital base.

- **Return on Equity (ROE)**

  Interpretation: ROE remained strong and improved slightly from 11.10% to 11.81%, indicating that the firm consistently delivered healthy returns to equity shareholders.

- **Earnings Per Share (EPS)**
  Interpretation: EPS rose from 14.39 in 2021 to 22.42 in 2025. This growth highlights NTPC's strong earnings potential and profitability expansion.
- **Earnings Yield**

  Interpretation: This declined from 13.50% in 2021 to 6.27% in 2025, suggesting either an increase in market price or slower EPS growth relative to market valuations, possibly reflecting higher investor confidence or market optimism.

- **Dividend Yield**
  Interpretation: Dividend yield dropped to 0% in 2025 from 5.35% in 2021. The zero yield may indicate retained earnings for future investment, though it may concern income-focused investors.
- **Dividend Payout Ratio**
  Interpretation: Similarly, the payout ratio fell to 0% in 2025, reinforcing the firm's emphasis on internal reinvestment over dividend distribution.
- **Price to Earnings Ratio (P/E)**
  Interpretation: The P/E ratio increased from 7.41 in 2021 to 15.95 in 2025, suggesting improved market valuation and investor expectations of future earnings growth.
- **Return on Investment (ROI)**
  Interpretation: ROI trends were identical to ROA, improving from 3.50% to 4.15%, reaffirming enhanced asset productivity and capital efficiency.
- **Current Assets Turnover**
  Interpretation: This ratio increased until 2023 and slightly declined thereafter, ending at 1.68 in 2025. This suggests a slight drop in efficiency of current asset utilization in the most recent year.
- **Fixed Assets Turnover**
  Interpretation: After peaking in 2023 at 0.53, it slightly declined to 0.51 in 2025, indicating minor volatility in fixed asset productivity, possibly due to capital additions or underutilized capacity.
- **Total Assets Turnover**
  Interpretation: The ratio remained low but steady, improving marginally from 0.28 in 2021 to 0.36 in 2025. It indicates moderate efficiency in using total assets to generate sales.

- **Inventory Turnover**
  Interpretation: The ratio improved significantly to 14.46 in 2023 but declined to 10.24 in 2025, showing variability in inventory usage efficiency, possibly due to changes in production or stockpiling.
- **Debtors Turnover**
  Interpretation: This ratio showed relative consistency around 5.5 to 6.7 times, suggesting stable receivables management and credit policy.
- **Degree of Total Leverage (DTL)**
  Interpretation: DTL remained stable in the 2.4–2.6 range, reflecting a consistent total risk exposure in relation to changes in sales and earnings.
- **Degree of Operating Leverage (DOL)**
  Interpretation: DOL remained around 1.2 to 1.5, indicating moderate operating lever- age and a balanced cost structure, with a low proportion of fixed costs.
- **Degree of Financial Leverage (DFL)**
  Interpretation: DFL declined from 2.05 in 2021 to 1.81 in 2025, reflecting reduced financial risk and improved interest burden management.
- **Overall Assessment**
  Interpretation: NTPC exhibits improving operational efficiency, declining leverage, and stable profitability. Strategic deleveraging, enhanced returns, and robust cost control have strengthened financial fundamentals. However, the absence of dividend payouts in 2025 and decreasing dividend yield may affect investor sentiment. The company appears to be reinvesting profits for future growth, with market confidence reflected in a rising P/E ratio.

B. Dabur India Limited

Dabur India Ratio Analysis						
Ratio Type	**Ratio Formula**	**2025**	**2024**	**2023**	**2022**	**2021**
Liquidity Ratios						
Current Ratio	= Current Assets/Current Liabilities	1.512689768	1.44503782	1.177183101	1.299259001	1.627686005
Quick Ratio	= (Current Assets - Inventory - Prepaid Expenses)/Current Liabilities	0.98006456	0.949679806	0.616374331	0.723985264	1.036626803
Leverage Ratios						
Debt-Equity Ratio	= Long Term Debt/Shareholders Equity	0.0281	0.0543	0.0333	0.0299	0.0175
Debt to Total Assets Ratio	= Total Debt/Total Assets	0.04497951	0.076578358	0.073164855	0.070629483	0.044552799
Proprietary Ratio	= Shareholders Equity/Total Assets	0.665383421	0.652417429	0.657171294	0.682264604	0.70650301
Interest Coverage Ratio	= EBIT/Interest	14.16721713	19.32863585	27.66002045	58.3888601	65.00097371
Profitability Ratios related to Sales						
Gross Profit Ratio	= Gross Profit (Sales-COGS) / Sales	49.952997%	47.398543%	45.411014%	49.414438%	51.194247%
Operating Profit Ratio	= EBIT/Sales	18.44%	19.35%	18.77%	20.70%	20.98%
Net Profit Ratio	= EAT/Sales	14.07%	14.86%	14.82%	15.99%	17.75%
Administrative Expenses Ratio	= Admin Expenses/Sales	6.88%	6.85%	5.55%	7.14%	8.22%
Selling Expenses Ratio	= Selling Expenses/Sales	6.88%	6.85%	5.55%	7.14%	8.22%
Operating Expenses Ratio	= (Admin Expenses + Selling Expenses) / Sales	6.88%	6.85%	5.55%	7.14%	8.22%
Operating Ratio	= (COGS+Operating Expenses) / Sales	56.92939%	59.44650%	60.14212%	57.73005%	57.02187%
Profitability Ratios related to Investment						
Return on Assets	= EAT/Total Assets	10.886%	12.188%	12.515%	14.172%	15.620%
Return on Capital Employed	= EBIT/Total Capital Employed	19.44%	21.45%	21.54%	25.15%	25.31%
Return on Equity	= EAT/Shareholders Equity	16.36%	18.68%	19.04%	20.77%	22.11%
Dabur India Ratio Analysis						
Ratio Type	**Ratio Formula**	**2025**	**2024**	**2023**	**2022**	**2021**
Profitability Ratios related to Equity Shares						
Earnings Per Share	= Net Profit Available to Equity Shareholders/Number of Equity Shares	9.97	10.40	9.64	9.85	9.59
Earnings Yield	= EPS/Market Price Per Share	1.96850%	1.98825%	1.76996%	1.83667%	1.77360%
Dividend Yield	= DPS/MPS	1.08589%	1.04176%	0.95430%	1.02574%	0.61980%
Dividend Payout Ratio	= DPS/EPS	55.16%	52.40%	53.92%	55.85%	34.95%
Price to Earnings Ratio	= MPS/EPS	50.80013735	50.29553127	56.49841751	54.44649615	56.3824337
Profitability Ratios related to Earning Power						
Return on Investment (ROI)	= EAT/Total Assets	10.8862%	12.1882%	12.5145%	14.1725%	15.6199%
Activity or Efficiency Ratios						
Current Assets Turnover	= Sales/Current Assets	1.923179017	2.183950687	2.713579323	2.522372847	1.998894458
Fixed Assets Turnover	= Sales/Fixed Assets	3.346953183	3.40572086	3.443269136	4.895438912	4.647334012
Total Assets Turnover	= Sales/Total Assets	0.773957201	0.820225648	0.844410251	0.886373349	0.880103769
Inventory Turnover	= Sales / Average Inventory	5.91610707	6.24703047	5.85932406	5.97351913	6.13170191
Debtors Turnover	= Sales / Average Debtors	14.05880641	14.19653556	15.42563382	18.03164615	13.88121878
Assessment of Risk						
DTL	= Contribution/EBT	1.597335032	1.575692403	1.440067606	1.412053375	1.266607802
DOL	= Contribution/EBIT	1.558786707	1.547555859	1.476419977	1.427334159	1.303638125
DFL	= EBIT/EBT	1.024729699	1.018181278	0.975378028	0.98929418	0.97159463

***Figure 2**: Ratio Analysis of Dabur India Limited from Excel Workings*

- **Current Ratio**
  Interpretation: The current ratio has moved from 1.627 in 2021 to 1.512 in 2025, with some fluctuations in between. The company has consistently maintained a current ratio above 1, reflecting its strong short-term liquidity position. Despite a slight decrease from 2021, the ratio remains healthy in 2025, indicating that Dabur is capable of meeting its short-term obligations comfortably.
- **Quick Ratio**
  Interpretation: The quick ratio increased from 1.036 in 2021 to 0.980 in 2025 after dipping to 0.616 in 2023. The quick ratio indicates an improving ability to meet short-term liabilities using its most liquid assets. While there was a concerning dip in 2023, the recovery by 2025 reflects better management of inventory and working capital efficiency.
- **Debt-Equity Ratio**
  Interpretation: The debt-equity ratio rose from 0.017 in 2021 to 0.028 in 2025, peaking at 0.054 in 2024. Dabur operates with minimal financial leverage. The ratio remains significantly below 1 throughout the period, signifying a conservative capital structure with low dependency on debt financing.
- **Debt to Total Assets Ratio**
  Interpretation: The ratio rose from 0.045 in 2021 to 0.077 in 2024 before dropping to 0.045 in 2025. Although the company's reliance on external debt increased briefly, it returned to its 2021 level in 2025. This demonstrates efficient debt management and minimal solvency risk.
- **Proprietary Ratio**
  Interpretation: The proprietary ratio declined from 0.707 in 2021 to 0.652 in 2024, before rising to 0.665 in 2025. A high proprietary ratio consistently above 0.65 shows a strong capital structure with a significant portion of assets financed by shareholders' equity, ensuring financial stability.
- **Interest Coverage Ratio**
  Interpretation: The interest coverage ratio dropped from 65.00 in 2021 to 14.17 in 2025. Despite the sharp decline, the ratio remains well above the acceptable benchmark of 3, indicating strong capability to service interest obligations. However, declining operating efficiency may warrant attention.
- **Gross Profit Ratio**
  Interpretation: Gross profit margin reduced from 51.19% in 2021 to 49.95% in 2025, after hitting a low of 45.41% in 2023. The company maintains a robust gross profit margin, suggesting efficient production and cost control. The recovery post-2023 shows improved pricing and cost strategies.
- **Operating Profit Ratio**
  Interpretation: Operating margin declined slightly from 20.98% in 2021 to 18.44% in 2025. Though still strong, the decrease may point toward rising operating expenses or pricing pressures. The company maintains operational profitability but should monitor cost efficiency.

- **Net Profit Ratio**
  Interpretation: Net profit margin declined from 17.75% in 2021 to 14.07% in 2025. The ratio reflects stable profitability, though a declining trend highlights increasing operational or tax-related costs. Dabur continues to generate good returns from its sales revenue.
- **Administrative Expenses Ratio**
  Interpretation: This ratio reduced from 8.22% in 2021 to 6.88% in 2025. Improved cost control over administrative expenses is evident, indicating better operational efficiency and streamlined internal processes.
- **Selling Expenses Ratio**
  Interpretation: Selling expenses ratio fell from 8.22% in 2021 to 6.88% in 2025. A reduction in selling expenses suggests improved marketing efficiency and cost-conscious promotion strategies.
- **Operating Expenses Ratio**
  Interpretation: Operating expenses ratio also fell from 8.22% in 2021 to 6.88% in 2025. Reduced operating expenses enhance profitability. Dabur demonstrates efficient cost management across core operations.
- **Operating Ratio**
  Interpretation: The operating ratio fluctuated, decreasing slightly from 57.02% in 2021 to 56.93% in 2025. A marginal decline indicates marginal improvement in operating efficiency. Lower ratios imply better cost control relative to revenue.
- **Return on Assets (ROA)**
  Interpretation: ROA dropped from 15.62% in 2021 to 10.89% in 2025. Although still respectable, the decline suggests reduced profitability of assets deployed. It may be due to increased asset base or margin compression.
- **Return on Capital Employed (ROCE)**
  Interpretation: ROCE declined from 25.31% in 2021 to 19.44% in 2025. The reduction signals declining efficiency in capital utilization, possibly due to increased capital investments or reduced EBIT growth.
- **Return on Equity (ROE)**
  Interpretation: ROE dropped from 22.11% in 2021 to 16.36% in 2025. While ROE remains attractive, the decreasing trend may concern equity investors regarding return generation from their capital.
- **Earnings Per Share (EPS)**
  Interpretation: EPS increased from 9.59 in 2021 to 9.97 in 2025. Stable and marginally growing EPS reflects consistent profitability, supporting long-term shareholder confidence.
- **Earnings Yield**
  Interpretation: Earnings yield moved slightly from 1.77% in 2021 to 1.97% in 2025. The increase indicates marginal improvement in investor returns per share price, though still relatively low, typical for growth companies.
- **Dividend Yield**
  Interpretation: Dividend yield rose from 0.62% in 2021 to 1.09% in 2025. The rising trend reflects enhanced shareholder rewards and a balanced dividend policy in line with profitability.

- **Dividend Payout Ratio**
  Interpretation: The payout ratio increased from 34.95% in 2021 to 55.16% in 2025. This shows Dabur's commitment to sharing profits with shareholders and a mature dividend policy.
- **Price to Earnings Ratio (P/E)**
  Interpretation: P/E declined slightly from 56.38 in 2021 to 50.80 in 2025. A decreasing P/E suggests either improved earnings or moderation in stock price growth, possibly making the stock relatively more attractive.
- **Return on Investment (ROI)**
  Interpretation: ROI dropped from 15.62% in 2021 to 10.89% in 2025. The downward trend reflects lower returns on deployed capital. Efficient investment strategies are needed to maintain performance.
- Current Assets Turnover
  Interpretation: The ratio moved from 1.999 in 2021 to 1.923 in 2025. A stable current assets turnover indicates consistent utilization of current assets to generate revenue.
- **Fixed Assets Turnover**
  Interpretation: The ratio declined from 4.65 in 2021 to 3.35 in 2025. Declining fixed asset efficiency may point to over-investment or underutilization of fixed assets in recent years.
- **Total Assets Turnover**
  Interpretation: It fell from 0.880 in 2021 to 0.774 in 2025. The company is generating slightly lower revenue per unit of total assets, hinting at reduced overall asset productivity.
- **Inventory Turnover**
  Interpretation: It decreased from 6.13 in 2021 to 5.92 in 2025. A marginal dip suggests slower inventory movement, which may affect liquidity if not monitored.
- **Debtors Turnover**
  Interpretation: The ratio remained stable, from 13.88 in 2021 to 14.06 in 2025. Stable turnover implies consistent credit policies and efficient collection practices from customers.
- **Degree of Total Leverage (DTL)**
  Interpretation: The DTL increased from 1.27 in 2021 to 1.60 in 2025. Rising DTL indicates that a small change in sales leads to a larger change in earnings, reflecting higher operational and financial leverage.
- **Degree of Operating Leverage (DOL)**
  Interpretation: DOL increased from 1.30 in 2021 to 1.56 in 2025. An increase in DOL suggests higher sensitivity of EBIT to changes in sales, highlighting fixed cost structures.
- **Degree of Financial Leverage (DFL)**
  Interpretation: DFL rose from 0.97 in 2021 to 1.02 in 2025. The ratio remains close to 1, indicating minimal financial risk and stable interest obligations.
- **Summary Interpretation**
  Interpretation: Dabur exhibits a fundamentally sound financial structure with strong liquidity, minimal leverage, and consistent profitability. The company maintains high gross margins and operating efficiency while exercising disciplined control over administrative and selling expenses. Return metrics have shown some decline, likely due to expansionary investments or market dynamics, but they remain healthy overall. Low debt reliance and

prudent capital allocation reflect conservative financial management. The company balances growth with shareholder returns through a rising dividend payout and stable EPS. While certain efficiency ratios and return metrics warrant closer observation, Dabur's financials suggest long-term sustainability and operational resilience.

C. HDFC Life Insurance

HDFC Life Insurance Ratio Analysis						
Ratio Type	**Ratio Formula**	**2025**	**2024**	**2023**	**2022**	**2021**
Liquidity Ratios						
Current Ratio	= Current Assets/Current Liabilities	1.03301657	0.91438113	0.86419931	0.9048715	0.77525667
Quick Ratio	= (Current Assets - Inventory - Prepaid Expenses)/Current Liabilities	1.03301657	0.91438113	0.86419931	0.9048715	0.77525667
Leverage Ratios						
Debt-Equity Ratio	= Long Term Debt/Shareholders Equity	0.18261753	0.06477413	0.07313324	0.03843687	0.05360575
Debt to Total Assets Ratio	= Total Debt/Total Assets	0.00844277	0.00313856	0.00383262	0.00260039	0.00333999
Proprietary Ratio	= Shareholders Equity/Total Assets	0.04623195	0.04845386	0.05240604	0.06765357	0.06230657
Interest Coverage Ratio	= EBIT/Interest	8.65181233	7.33645794	0.6318408	28.9741885	73.8079219
Profitability Ratios related to Sales						
Gross Profit Ratio	= Gross Profit (Sales-COGS) / Sales	8.22%	7.88%	14.03%	10.90%	16.69%
Operating Profit Ratio	= EBIT/Sales	1.1378%	0.5784%	0.0742%	1.1037%	1.9030%
Net Profit Ratio	= EAT/Sales	1.8683%	1.5511%	1.9489%	1.9768%	1.9038%
Administrative Expenses Ratio	= Admin Expenses/Sales	1.0766%	1.7442%	3.5172%	2.8167%	1.9729%
Selling Expenses Ratio	= Selling Expenses/Sales	1.0766%	1.7442%	3.5172%	2.8167%	1.9729%
Operating Expenses Ratio	= (Admin Expenses + Selling Expenses) / Sales	1.0766%	1.7442%	3.5172%	2.8167%	1.9729%
Operating Ratio	= (COGS+Operating Expenses) / Sales	92.85%	93.87%	89.49%	91.92%	85.28%
Profitability Ratios related to Investment						
Return on Assets	= EAT/Total Assets	0.518%	0.520%	0.552%	0.575%	0.758%
Return on Capital Employed	= EBIT/Total Capital Employed	0.325%	0.200%	0.022%	0.331%	0.786%
Return on Equity	= EAT/Shareholders Equity	11.21%	10.73%	10.53%	8.50%	12.16%

HDFC Life Insurance Ratio Analysis						
Ratio Type	**Ratio Formula**	**2025**	**2024**	**2023**	**2022**	**2021**
Profitability Ratios related to Equity Shares						
Earnings Per Share	= Net Profit Available to Equity Shareholders/Number of Equity Shares	8.41	7.32	6.37	6.28	6.73
Earnings Yield	= EPS/Market Price Per Share	1.23%	1.16%	1.28%	1.14%	0.97%
Dividend Yield	= DPS/MPS	0.29%	0.30%	0.33%	0.35%	0.00%
Dividend Payout Ratio	= DPS/EPS	23.76%	25.95%	26.25%	30.78%	0.00%
Price to Earnings Ratio	= MPS/EPS	81.526815	86.5458161	78.4180492	87.6217937	102.616143
Profitability Ratios related to Earning Power						
Return on Investment (ROI)	= EAT/Total Assets	0.5182%	0.5200%	0.5520%	0.5751%	0.7575%
Activity or Efficiency Ratios						
Current Assets Turnover	= Sales/Current Assets	9.66716504	12.5516274	9.9421206	10.6772157	14.1417626
Fixed Assets Turnover	= Sales/Fixed Assets	159.253598	241.427868	183.605523	181.426093	209.307361
Total Assets Turnover	= Sales/Total Assets	0.28	0.34	0.28	0.29	0.40
Inventory Turnover	= Sales / Average Inventory	0.000000	0.000000	0.000000	0.000000	0.000000
Debtors Turnover	= Sales / Average Debtors	0.000000	0.000000	0.000000	0.000000	0.000000
Assessment of Risk						
DTL	= Contribution/EBT	0.85502504	0.60063646	0.03612661	0.49814759	0.83521423
DOL	= Contribution/EBIT	1.000000	1.000000	1.000000	1.000000	1.000000
DFL	= EBIT/EBT	0.60898378	0.37290989	0.03805508	0.5583339	0.99956645

Figure 3: *Ratio Analysis of HDFC Life Insurance from Excel Workings*

- **Current Ratio**
  Interpretation: The current ratio has shown a gradual improvement from 0.775 in 2021 to 1.033 in 2025. The company's ability to meet short-term obligations has im- proved over the years, surpassing the benchmark of 1 in 2025. This indicates enhanced liquidity and better short-term financial health.
- **Quick Ratio**
  Interpretation: The quick ratio has followed the same pattern as the current ratio, increasing from 0.775 in 2021 to 1.033 in 2025. The consistent rise in the quick ratio reflects HDFC Life's increased ability to meet short-term liabilities using its most liquid assets, showcasing stronger financial flexibility.
- **Debt-Equity Ratio**
  Interpretation: The debt-equity ratio increased from 0.054 in 2021 to 0.183 in 2025. There has been a rise in the reliance on long-term debt compared to shareholders' equity. Although

still relatively low, the increase implies a slightly higher financial leverage position over the years.

- **Debt to Total Assets Ratio**
  Interpretation: This ratio rose from 0.003 in 2021 to 0.008 in 2025. The company has marginally increased its total debt relative to assets. The ratio remains low, indicating that HDFC Life maintains a conservative approach to debt financing.
- **Proprietary Ratio**
  Interpretation: This ratio declined from 0.062 in 2021 to 0.046 in 2025. The declining proprietary ratio indicates a reduced share of shareholders' funds in the total assets, which may slightly impact long-term financial stability.
- **Interest Coverage Ratio**
  Interpretation: There was a decline from 73.81 in 2021 to 0.63 in 2023, but a recovery to 8.65 in 2025. The interest coverage ratio rebounded sharply, signifying improved earnings before interest and tax. However, the sharp drop in 2023 highlights a period of poor operational performance.
- **Gross Profit Ratio**
  Interpretation: This ratio decreased from 16.69% in 2021 to 8.22% in 2025. A declining gross profit ratio suggests rising cost of goods sold or pressure on pricing power, potentially indicating narrowing margins.
- **Operating Profit Ratio**
  Interpretation: It decreased from 1.90% in 2021 to 1.14% in 2025, with a major dip in 2023. Although there's a recovery in the recent years, the lower operating profit ratio signals operational inefficiencies or increased operating costs.
- **Net Profit Ratio**
  Interpretation: Relatively stable around 1.9% from 2021 to 2025, with minor fluctuations. The company has maintained stable net profitability, reflecting consistent bottom-line performance relative to revenue.
- **Administrative Expenses Ratio**
  Interpretation: It declined from 1.97% in 2021 to 1.08% in 2025. A steady reduction in administrative expenses indicates better control over indirect costs and enhanced operational efficiency.
- **Selling Expenses Ratio**
  Interpretation: Same as administrative expenses, it declined over the years to 1.08%. This suggests improved efficiency in marketing and distribution spending, contributing positively to cost management.
- **Operating Expenses Ratio**
  Interpretation: Mirroring admin and selling expenses, the ratio declined to 1.08% by 2025. The company has managed to streamline its operational expenses over the years, boosting cost control and supporting profitability.
- **Operating Ratio**
  Interpretation: It increased from 85.28% in 2021 to 92.85% in 2025. A rising operating ratio shows an increase in total operating costs as a percentage of sales, reducing the margin available for profits.

- **Return on Assets (ROA)**
  Interpretation: ROA decreased slightly from 0.76% in 2021 to 0.52% in 2025. The marginal decline suggests the company is generating slightly lower returns on its total asset base over time.
- **Return on Capital Employed (ROCE)**
  Interpretation: ROCE declined from 0.79% in 2021 to 0.33% in 2025. There is a downward trend in the efficiency of capital utilization, implying that returns on funds employed are diminishing.
- **Return on Equity (ROE)**
  Interpretation: ROE showed a dip in 2022 (8.5%) but recovered to 11.21% in 2025. The upward trend in ROE indicates stronger returns to equity shareholders, suggesting improving overall profitability and efficient use of equity capital.
- **Earnings Per Share (EPS)**
  Interpretation: EPS grew from 6.73 in 2021 to 8.41 in 2025. This growth in EPS signifies improved earnings attributable to each share, benefiting shareholders and reflecting financial strength.
- **Earnings Yield**
  Interpretation: It fluctuated but improved marginally from 0.97% in 2021 to 1.23% in 2025. The slight rise in earnings yield implies marginal improvement in the return investors are receiving for each rupee invested.
- **Dividend Yield**
  Interpretation: Increased from 0% in 2021 to 0.29% in 2025. The reintroduction and increase in dividend yield indicate the company's commitment to returning value to shareholders.
- **Dividend Payout Ratio**
  Interpretation: Declined from 30.78% in 2022 to 23.76% in 2025. This suggests that the company is retaining a greater portion of its earnings for reinvestment or reserves, reflecting a long-term growth strategy.
- **Price to Earnings Ratio (P/E)**
  Interpretation: Declined from 102.62 in 2021 to 81.53 in 2025. A falling P/E ratio could indicate either undervaluation of stock or reduced investor confidence in future growth.
- **Return on Investment (ROI)**
  Interpretation: Declined from 0.76% in 2021 to 0.52% in 2025. This trend highlights reduced effectiveness in utilizing assets to generate profits.
- **Current Assets Turnover**
  Interpretation: This ratio dropped from 14.14 in 2021 to 9.67 in 2025. The fall shows a decline in efficiency of current asset utilization in generating sales.
- **Fixed Assets Turnover**
  Interpretation: Decreased from 209.31 in 2021 to 159.25 in 2025. This indicates that the company is generating lower sales from its fixed asset base, possibly due to over-investment or underutilization.

- **Total Assets Turnover**
  Interpretation: Declined from 0.40 in 2021 to 0.28 in 2025. The overall efficiency in using total assets to generate revenue has reduced, signaling lower operational efficiency.
- **Inventory Turnover**
  Interpretation: Zero across all years. As an insurance company, HDFC Life holds minimal or no inventory, rendering this ratio inapplicable.
- **Debtors Turnover**
  Interpretation: Zero across all years. This is expected due to the nature of the insurance business model, where receivables are typically negligible.
- **Degree of Total Leverage (DTL)**
  Interpretation: Stable between 0.49 and 0.85, with minor fluctuations. The relatively consistent DTL suggests moderate sensitivity of the company's earnings to changes in sales.
- **Degree of Operating Leverage (DOL)**
  Interpretation: Constant at 1.00 over all five years. A constant DOL of 1 indicates that changes in sales do not disproportionately impact EBIT, showing stable cost structure.
- **Degree of Financial Leverage (DFL)**
  Interpretation: Fluctuated from 0.56 to 1.00, but stood at 0.61 in 2025. The relatively low DFL indicates low dependence on debt for financing, minimizing financial risk.
- **Summary Interpretation**
  Interpretation: HDFC Life Insurance exhibits a balanced financial profile with improving liquidity, controlled leverage, and consistent profitability. Operational and asset efficiency ratios show slight declines, indicating areas for improvement in re- source utilization. The company has maintained shareholder value through a growing EPS and ROE, along with steady dividend policies. Risk-related ratios remain within manageable levels, affirming financial stability.

D. Adani Enterprises

Adani Enterprises Ratio Analysis						
Ratio Type	**Ratio Formula**	**2025**	**2024**	**2023**	**2022**	**2021**
Liquidity Ratios						
Current Ratio	= Current Assets/Current Liabilities	0.8598937	0.8255027	0.8285536	0.7057137	0.9292569
Quick Ratio	= (Current Assets - Inventory - Prepaid Expenses)/Current Liabilities	0.6427829	0.6130391	0.6741434	0.5509061	0.8474698
Leverage Ratios						
Debt-Equity Ratio	= Long Term Debt/Shareholders Equity	1.3546089	1.2393533	1.0224679	0.9623862	0.5550171
Debt to Total Assets Ratio	= Total Debt/Total Assets	0.384769	0.3118478	0.2708333	0.4031416	0.296136
Proprietary Ratio	= Shareholders Equity/Total Assets	0.2406937	0.2194647	0.2252766	0.212426	0.3322544
Interest Coverage Ratio	= EBIT/Interest	2.384072	2.4977715	2.2236837	1.4700619	1.8194066
Profitability Ratios related to Sales						
Gross Profit Ratio	= Gross Profit (Sales-COGS) / Sales	50.23%	49.24%	24.72%	24.20%	22.61%
Operating Profit Ratio	= EBIT/Sales	14.559%	11.799%	6.445%	5.349%	6.336%
Net Profit Ratio	= EAT/Sales	6.7330%	3.4169%	1.6126%	0.6687%	1.5762%
Administrative Expenses Ratio	= Admin Expenses/Sales	0.8408%	0.7041%	0.0000%	0.2129%	0.4222%
Selling Expenses Ratio	= Selling Expenses/Sales	0.8408%	0.7041%	0.0000%	0.2129%	0.4222%
Operating Expenses Ratio	= (Admin Expenses + Selling Expenses) / Sales	0.8408%	0.7041%	0.0000%	0.2129%	0.4222%
Operating Ratio	= (COGS+Operating Expenses) / Sales	50.61%	51.46%	75.28%	76.01%	77.81%
Profitability Ratios related to Investment						
Return on Assets	= EAT/Total Assets	3.327%	2.050%	1.561%	0.456%	1.207%
Return on Capital Employed	= EBIT/Total Capital Employed	4.372%	2.838%	2.285%	0.802%	2.066%
Return on Equity	= EAT/Shareholders Equity	13.82%	9.34%	6.93%	2.15%	3.63%

Adani Enterprises Ratio Analysis						
Ratio Type	**Ratio Formula**	**2025**	**2024**	**2023**	**2022**	**2021**
Profitability Ratios related to Equity Shares						
Earnings Per Share	= Net Profit Available to Equity Shareholders/Number of Equity Shares	57.11	28.90	19.38	4.22	5.67
Earnings Yield	= EPS/Market Price Per Share	2.47%	0.90%	1.11%	0.21%	0.51%
Dividend Yield	= DPS/MPS	0.0554%	0.0375%	0.000%	0.0489%	0.00%
Dividend Payout Ratio	= DPS/EPS	2.248%	4.152%	0.000%	23.691%	0.000%
Price to Earnings Ratio	= MPS/EPS	40.551401	110.62539	90.338035	484.16231	195.39628
Profitability Ratios related to Earning Power						
Return on Investment (ROI)	= EAT/Total Assets	3.327%	2.050%	1.561%	0.456%	1.207%
Activity or Efficiency Ratios						
Current Assets Turnover	= Sales/Current Assets	2.4026512	2.6158639	3.6899616	2.2433125	1.9804897
Fixed Assets Turnover	= Sales/Fixed Assets	0.7666601	0.9630754	1.7118116	1.300825	2.0263916
Total Assets Turnover	= Sales/Total Assets	0.4940797	0.5998872	0.968124	0.6821939	0.7655875
Inventory Turnover	= Sales / Average Inventory	9.9012352	11.755137	19.987518	16.247532	18.306727
Debtors Turnover	= Sales / Average Debtors	10.087355	8.6298935	10.430413	5.4034335	3.1467107
Assessment of Risk						
DTL	= Contribution/EBT	4.0693544	8.5023582	9.8035447	7.6666035	6.4306150
DOL	= Contribution/EBIT	2.9800906	3.6809697	3.6090408	1.96067	2.7779446
DFL	= EBIT/EBT	1.3655137	2.3098148	2.7163851	3.9101957	2.3148824

Figure 4: *Ratio Analysis of Adani Enterprises from Excel Workings*

- **Current Ratio**
  Interpretation: The current ratio has declined from 0.93 in 2021 to 0.86 in 2025, with slight fluctuations over the years. The company's ability to meet short-term liabilities remains below the ideal benchmark of 1 throughout the five-year period. Although there is a marginal improvement in 2025 compared to 2022, the overall low ratio reflects a tight liquidity position, which may pose challenges in covering current obligations without relying on additional financing.
- **Quick Ratio**
  Interpretation: The quick ratio dropped from 0.85 in 2021 to 0.64 in 2025. The declining trend suggests a weakening ability to meet short-term liabilities using liquid assets excluding inventories. A quick ratio consistently below 1 indicates potential liquidity constraints and reliance on inventory clearance or external funding to meet immediate liabilities.
- **Debt-Equity Ratio**
  Interpretation: The ratio increased from 0.56 in 2021 to 1.35 in 2025. This significant rise highlights an increased dependence on long-term debt relative to shareholders' equity. A debt-equity ratio above 1 in recent years indicates aggressive leverage and potentially higher financial risk.
- **Debt to Total Assets Ratio**
  Interpretation: The ratio moved from 0.30 in 2021 to 0.38 in 2025, peaking at 0.40 in 2022. The proportion of total assets funded through debt has steadily increased. While still under 50%, the rising trend suggests growing financial obligations, warranting cautious debt management.
- **Proprietary Ratio**
  Interpretation: The proprietary ratio fell from 0.33 in 2021 to 0.24 in 2025. This decline indicates that the proportion of assets financed by shareholders has reduced over time, reflecting an increasing reliance on external funding and a potentially weakened equity base.
- **Interest Coverage Ratio**
  Interpretation: The ratio improved from 1.82 in 2021 to 2.38 in 2025, with a dip in 2022. The company has enhanced its capacity to service interest obligations, as evidenced by the

rising coverage ratio. However, the ratio remains relatively low, suggesting room for strengthening operational efficiency and profit margins.

- **Gross Profit Ratio**
  Interpretation: The ratio increased from 22.61% in 2021 to 50.23% in 2025. This sharp rise reflects enhanced pricing strategies, cost control, or favorable market conditions. A strong gross margin indicates increased efficiency in core operations.
- **Operating Profit Ratio**
  Interpretation: The ratio improved from 6.34% in 2021 to 14.56% in 2025. The consistent rise indicates stronger operational performance and better control over operating costs, translating into higher profitability from core business activities.
- **Net Profit Ratio**
  Interpretation: The net margin rose from 1.58% in 2021 to 6.73% in 2025. This improvement signals effective cost management and stronger bottom-line performance, though absolute margins remain modest in comparison to gross margins.
- **Administrative Expenses Ratio**
  Interpretation: This ratio decreased slightly from 0.42% in 2021 to 0.84% in 2025, with 0% recorded in 2023. The fluctuation indicates inconsistent administrative cost allocation. The recent increase may suggest growing overheads that need monitoring.
- **Selling Expenses Ratio**
  Interpretation: The ratio mirrors administrative expenses, increasing from 0.42% in 2021 to 0.84% in 2025. This parallel trend may indicate bundled reporting or growing marketing efforts. Rising selling costs should be evaluated against revenue growth effectiveness.
- **Operating Expenses Ratio**
  Interpretation: The ratio increased from 0.42% in 2021 to 0.84% in 2025. The rise in total operating expenses relative to sales is slight but reflects cost expansion. Monitoring efficiency across administrative and selling functions is advised.
- **Operating Ratio**
  Interpretation: The operating ratio decreased from 77.81% in 2021 to 50.61% in 2025. A significant decline indicates improved operational efficiency and reduced expenditure per unit of sales. A lower ratio is favorable for profitability.
- **Return on Assets (ROA)**
  Interpretation: ROA improved from 1.21% in 2021 to 3.33% in 2025. The company's asset utilization has steadily improved, resulting in enhanced profitability per unit of total assets.
- **Return on Capital Employed (ROCE)**
  Interpretation: ROCE rose from 2.07% in 2021 to 4.37% in 2025. The rising ROCE suggests more efficient use of capital in generating EBIT, reflecting better performance and strategic capital deployment.
- **Return on Equity (ROE)**
  Interpretation: ROE increased from 3.63% in 2021 to 13.82% in 2025. The growth indicates strong earnings generation relative to shareholder investments, reflecting high equity efficiency and improved value creation.

- **Earnings Per Share (EPS)**
  Interpretation: EPS grew significantly from 5.67 in 2021 to 57.11 in 2025. The sharp increase highlights enhanced profitability and earnings growth per share, which may support shareholder value.
- **Earnings Yield**
  Interpretation: The yield rose from 0.51% in 2021 to 2.47% in 2025. Improved returns relative to market price suggest better investor earnings expectations, though still modest in absolute terms.
- **Dividend Yield**
  Interpretation: The yield increased slightly from 0.00% in 2021 to 0.055% in 2025. The dividend return remains negligible, indicating the company may be reinvesting profits for growth rather than distributing to shareholders.
- **Dividend Payout Ratio**
  Interpretation: The payout ratio decreased from 23.69% in 2022 to 2.25% in 2025. A declining payout ratio reflects retained earnings strategy for internal growth. Share- holders may expect lower cash returns.
- **Price to Earnings Ratio (P/E)**
  Interpretation: The P/E ratio declined from 195.40 in 2021 to 40.55 in 2025. This indicates that the stock valuation has become more aligned with earnings, possibly reflecting improved market efficiency or reduced speculative premium.
- **Return on Investment (ROI)**
  Interpretation: ROI improved from 1.21% in 2021 to 3.33% in 2025. This trend reflects better efficiency in utilizing total assets to generate profits, aligned with gains in ROA.
- **Current Assets Turnover**
  Interpretation: The ratio increased from 1.98 in 2021 to 2.40 in 2025. Improved turnover indicates efficient utilization of current assets in generating revenue.
- **Fixed Assets Turnover**
  Interpretation: This ratio fell from 2.03 in 2021 to 0.77 in 2025. The decline suggests underutilization of fixed assets or increasing investments in fixed assets without proportional sales growth.
- **Total Assets Turnover**
  Interpretation: The ratio dropped from 0.77 in 2021 to 0.49 in 2025. Lower turnover indicates less effective use of total assets in generating revenue, potentially reflecting inefficiencies or asset buildup.
- **Inventory Turnover**
  Interpretation: The ratio declined from 18.31 in 2021 to 9.90 in 2025. Slower inventory turnover implies higher inventory holding, which may impact working capital and liquidity.
- **Debtors Turnover**
  Interpretation: The ratio rose from 3.15 in 2021 to 10.09 in 2025. Faster collection efficiency and improved credit management are evident, contributing positively to cash flows.

- **Degree of Total Leverage (DTL)**
  Interpretation: DTL increased from 6.43 in 2021 to 4.07 in 2025, peaking at 9.80 in 2023. The high values reflect sensitivity of earnings to changes in sales, indicating elevated business risk, though slightly moderating in 2025.
- **Degree of Operating Leverage (DOL)**
  Interpretation: DOL fell from 2.78 in 2021 to 2.98 in 2025, with a peak of 3.68 in 2023. The firm maintains a high degree of operating leverage, implying significant fixed costs and earnings sensitivity to sales fluctuations.
- **Degree of Financial Leverage (DFL)**
  Interpretation: DFL declined from 2.31 in 2021 to 1.37 in 2025. The lower financial leverage in recent years indicates reduced dependence on debt for earnings amplification, suggesting controlled financial risk.
- **Summary Interpretation**
  Interpretation: The overall ratio analysis of Adani Enterprises over the five-year period indicates a company undergoing significant financial and operational transformation. The liquidity position, though still below the ideal benchmark, has shown marginal improvement, reflecting cautious progress in managing current obligations. Leverage ratios highlight a growing reliance on debt, suggesting aggressive expansion strategies that increase financial risk but are partly balanced by improving interest coverage. Profitability ratios—especially those related to sales and investments—demonstrate a strong upward trend, underlining improved cost management, better margins, and increased returns to shareholders. The surge in earnings per share and return on equity reinforces this positive trajectory. However, volatility in dividend-related metrics reflects inconsistent shareholder distribution policies, possibly due to reinvestment priorities. Activity ratios reveal efficient utilization of assets, particularly in inventory and receivables management, though fixed asset turnover shows strain due to heavy capital investments. The risk assessment ratios (DTL, DOL, DFL) suggest that while the company faces high operating and financial leverage, its earnings sensitivity is stabilizing. In conclusion, Adani Enterprises is demonstrating improving operational efficiency, rising profitability, and strategic expansion. However, the increasing lever- age and modest liquidity warrant cautious monitoring to ensure long-term financial sustainability.

E. Hindalco Industries

Hindalco Industries Ratio Analysis

Ratio Type	Ratio Formula	2025	2024	2023	2022	2021
Liquidity Ratios						
Current Ratio	= Current Assets/Current Liabilities	1.556245803	1.387390271	1.477264833	1.275257852	1.394593034
Quick Ratio	= (Current Assets - Inventory - Prepaid Expenses)/Current Liabilities	0.811374321	0.699752321	0.766710224	0.699597531	0.76217186
Leverage Ratios						
Debt-Equity Ratio	= Long Term Debt/Shareholders Equity	0.454429346	0.446524467	0.542541297	0.660403903	0.886605841
Debt to Total Assets Ratio	= Total Debt/Total Assets	0.232831186	0.235012311	0.259477709	0.283486206	0.342721891
Proprietary Ratio	= Shareholders Equity/Total Assets	0.465087165	0.457692092	0.421685193	0.350516897	0.350708227
Interest Coverage Ratio	= EBIT/Interest	9.30242761	6.187662001	6.216675809	7.523089172	4.691278759
Profitability Ratios related to Sales						
Gross Profit Ratio	= Gross Profit (Sales-COGS) / Sales	41.60%	37.64%	35.42%	39.53%	43.99%
Operating Profit Ratio	= EBIT/Sales	13.34%	11.05%	10.15%	14.53%	13.29%
Net Profit Ratio	= EAT/Sales	6.708%	4.701%	4.520%	7.036%	2.635%
Administrative Expenses Ratio	= Admin Expenses/Sales	0	0	0	0	0
Selling Expenses Ratio	= Selling Expenses/Sales	0	0	0	0	0
Operating Expenses Ratio	= (Admin Expenses + Selling Expenses) / Sales	0	0	0	0	0
Operating Ratio	= (COGS+Operating Expenses) / Sales	58.40%	62.36%	64.58%	60.47%	56.01%
Profitability Ratios related to Investment						
Return on Assets	= EAT/Total Assets	6.015%	4.378%	4.487%	6.153%	1.833%
Return on Capital Employed	= EBIT/Total Capital Employed	7.981%	5.884%	6.138%	9.414%	2.463%
Return on Equity	= EAT/Shareholders Equity	12.93%	9.57%	10.64%	17.55%	5.23%

Hindalco Industries Ratio Analysis						
Ratio Type	Ratio Formula	2025	2024	2023	2022	2021
Profitability Ratios related to Equity Shares						
Earnings Per Share	= Net Profit Available to Equity Shareholders/Number of Equity Shares	72.07	45.73	45.35	61.70	15.63
Earnings Yield	= EPS/Market Price Per Share	10.56%	8.16%	11.19%	10.81%	4.46%
Dividend Yield	= DPS/MPS	0.00%	0.00%	0.99%	0.53%	0.28%
Dividend Payout Ratio	= DPS/EPS	0.00%	0.00%	8.82%	4.86%	6.38%
Price to Earnings Ratio	= MPS/EPS	9.469585599	12.25012312	8.937702303	9.246464799	22.40763146
Profitability Ratios related to Earning Power						
Return on Investment (ROI)	= EAT/Total Assets	6.015%	4.378%	4.487%	6.153%	1.833%
Activity or Efficiency Ratios						
Current Assets Turnover	= Sales/Current Assets	2.33913632	2.622712313	2.499154639	1.979430299	1.95163246
Fixed Assets Turnover	= Sales/Fixed Assets	2.03373412	2.14669688	2.41088344	2.22026316	1.51438832
Total Assets Turnover	= Sales/Total Assets	0.896631841	0.93124399	0.99281638	0.874460912	0.695760125
Inventory Turnover	= Sales / Average Inventory	5.322799147	5.156070192	5.105202365	5.191121875	4.975684234
Debtors Turnover	= Sales / Average Debtors	13.16275733	13.24189098	11.97114508	11.46225944	11.83509684
Assessment of Risk						
DTL	= Contribution/EBT	2.749977613	4.641970021	3.601277634	1.982891234	4.297125053
DOL	= Contribution/EBIT	1.931080019	2.724279491	2.213579811	1.418739196	1.730269161
DFL	= EBIT/EBT	1.424061968	1.703925767	1.626902096	1.39764323	2.483500921

***Figure 5**: Ratio Analysis of Hindalco Industries from Excel Workings*

- **Current Ratio**
  Interpretation: The current ratio has declined slightly from 1.394 in 2021 to 1.556 in 2025, with some fluctuations in between. The company has maintained a current ratio above 1 throughout the five years, indicating a consistent ability to meet short-term obligations. The increase in 2025 further strengthens the firm's short-term financial position.
- **Quick Ratio**
  Interpretation: The quick ratio moved from 0.762 in 2021 to 0.811 in 2025, after a dip in 2022 and 2024. Although below the ideal benchmark of 1, the gradual increase in 2025 suggests improving liquidity through better management of current assets excluding inventory.
- **Debt-Equity Ratio**
  Interpretation: The ratio has decreased from 0.887 in 2021 to 0.454 in 2025. Hindalco has significantly reduced its dependence on long-term debt relative to equity, indicating a more conservative capital structure and lower financial risk.
- **Debt to Total Assets Ratio**
  Interpretation: This ratio has improved from 0.343 in 2021 to 0.233 in 2025. The company's reliance on debt to finance its assets has declined, reflecting prudent debt management and a stronger asset base.
- **Proprietary Ratio**
  Interpretation: The ratio increased from 0.351 in 2021 to 0.465 in 2025. A higher proportion of assets is financed by shareholders' equity, which reflects increased financial stability and investor confidence.
- **Interest Coverage Ratio**
  Interpretation: The ratio improved significantly from 4.691 in 2021 to 9.302 in 2025. The company has enhanced its ability to cover interest obligations, reflecting improved operational earnings and reduced interest burden.
- **Gross Profit Ratio**
  Interpretation: Gross profit margin decreased from 43.99% in 2021 to 41.60% in 2025, after fluctuations. Despite slight variations, the gross margin remains strong, indicating efficient cost control in production over the years.

- **Operating Profit Ratio**
  Interpretation: The operating profit ratio rose from 13.29% in 2021 to 13.34% in 2025. The company has marginally improved its operational efficiency, translating sales into operating income more effectively.
- **Net Profit Ratio**
  Interpretation: Net profit ratio improved from 2.635% in 2021 to 6.708% in 2025. The profitability after tax has shown a consistent upward trend, indicating better control over costs and enhanced bottom-line performance.
- **Administrative Expenses Ratio**
  Interpretation: This ratio is reported as zero across all years. Administrative expenses were either not disclosed or not separately classified. Thus, the ratio does not offer insights.
- **Selling Expenses Ratio**
  Interpretation: This ratio remains at zero throughout the period. Selling expenses were likely bundled under general costs or not reported separately, limiting the assessment of sales effort costs.
- **Operating Expenses Ratio**
  Interpretation: The ratio is shown as zero for all years. Absence of reported operating expenses may indicate consolidated expense reporting, making it difficult to assess overhead trends.
- **Operating Ratio**
  Interpretation: The operating ratio rose from 56.01% in 2021 to 58.40% in 2025, with fluctuations. A moderate rise indicates slightly higher operating expenses as a percentage of sales, but remains within efficient levels.
- **Return on Assets (ROA)**
  Interpretation: ROA increased from 1.833% in 2021 to 6.015% in 2025. The company has improved its efficiency in using assets to generate profits, reflecting enhanced operational performance.
- **Return on Capital Employed (ROCE)**
  Interpretation: ROCE rose from 2.463% in 2021 to 7.981% in 2025. Capital employed is being utilized more effectively to generate earnings before interest and tax, showing efficient capital management.
- **Return on Equity (ROE)**
  Interpretation: ROE increased from 5.23% in 2021 to 12.93% in 2025. Shareholders' returns have more than doubled, indicating stronger earnings growth and improved equity utilization.
- **Earnings Per Share (EPS)**
  Interpretation: EPS surged from 15.63 in 2021 to 72.07 in 2025. A consistent and strong rise in EPS reflects increased profitability and value generation for shareholders.
- **Earnings Yield**
  Interpretation: This ratio rose from 4.46% in 2021 to 10.56% in 2025. The company is offering higher earnings per market price of the share, making it potentially attractive to investors.

- **Dividend Yield**
  Interpretation: Dividend yield remained negligible or zero from 2021 to 2025. Low or no dividends indicate retained earnings strategy, possibly for reinvestment in growth and expansion activities.
- **Dividend Payout Ratio**
  Interpretation: The ratio declined to 0% in 2025 from 6.38% in 2021. The company has adopted a conservative dividend policy, choosing to retain earnings instead of distributing them to shareholders.
- **Price to Earnings Ratio (P/E)**
  Interpretation: P/E fell from 22.41 in 2021 to 9.47 in 2025. Lower P/E indicates the stock has become more reasonably priced relative to its earnings, potentially reflecting improved investor sentiment and earnings performance.
- **Return on Investment (ROI)**
  Interpretation: ROI increased from 1.833% in 2021 to 6.015% in 2025. The company has become more efficient at generating net returns from its total investments, reflecting better capital allocation.
- **Current Assets Turnover**
  Interpretation: This ratio rose from 1.952 in 2021 to 2.339 in 2025. Current assets are being utilized more effectively to generate revenue, showing improved operational efficiency.
- **Fixed Assets Turnover**
  Interpretation: The ratio increased from 1.514 in 2021 to 2.034 in 2025. The company has significantly improved its usage of fixed assets to generate sales, indicating better asset productivity.
- **Total Assets Turnover**
  Interpretation: The ratio improved from 0.696 in 2021 to 0.897 in 2025. Overall asset utilization has strengthened, contributing to higher revenue generation relative to asset base.
- **Inventory Turnover**
  Interpretation: The ratio increased from 4.976 in 2021 to 5.323 in 2025. Inventory is being turned over more frequently, reflecting better inventory management and faster conversion to sales.
- **Debtors Turnover**
  Interpretation: The ratio improved from 11.835 in 2021 to 13.163 in 2025. Receivables are being collected more quickly, suggesting strong credit management and cash flow efficiency.
- **Degree of Total Leverage (DTL)**
  Interpretation: DTL moved from 4.297 in 2021 to 2.750 in 2025, with some volatility. A decrease in DTL indicates reduced sensitivity of earnings to changes in sales, leading to more stable profitability.

- **Degree of Operating Leverage (DOL)**
  Interpretation: DOL decreased from 1.730 in 2021 to 1.931 in 2025. The company's fixed operating cost structure has become more flexible, reducing earnings volatility from sales changes.
- **Degree of Financial Leverage (DFL)**
  Interpretation: DFL declined from 2.484 in 2021 to 1.424 in 2025. Lower DFL implies reduced financial risk and dependence on debt for profit generation, improving financial stability.
- **Summary Interpretation**
  Interpretation: The ratio analysis of Hindalco Industries over the five-year period presents a picture of consistent financial strengthening and operational improvement. Liquidity has remained stable with a positive trend, while leverage ratios show a notable decline in debt dependency, enhancing financial soundness. Profitability metrics, particularly net profit ratio, ROE, and EPS, have shown substantial growth, indicating improved earnings capacity and value creation for shareholders. Despite a conservative dividend policy, retained earnings are likely being channeled into strategic investments, as reflected in stronger returns and efficient asset utilization. Risk metrics show a reduction in both operating and financial leverage, contributing to greater earnings stability. Overall, Hindalco exhibits a positive financial trajectory backed by prudent financial management, growing profitability, and efficient resource utilization.

The financial analysis of NTPC Limited, Dabur India Ltd., HDFC Life Insurance, Adani Enterprises, and Hindalco Industries reveals diverse financial trajectories reflective of their respective industries and strategic priorities. NTPC demonstrates stable growth with improved operational efficiency and reduced leverage, though its zero-dividend payout in 2025 may impact investor sentiment. Dabur maintains a conservative financial structure with strong liquidity and steady profitability, but declining return metrics suggest a need for enhanced efficiency. HDFC Life Insurance shows balanced growth with improving liquidity and stable profitability, though operational efficiency requires attention. Adani Enterprises exhibits aggressive expansion with significant improvements in profitability, tempered by high leverage and liquidity challenges. Hindalco stands out with robust profitability growth, reduced debt dependency, and efficient asset utilization, positioning it strongly for future growth. Collectively, these companies demonstrate resilience and strategic focus, with varying degrees of financial stability and growth potential.

2.4. Key Trends and Interpretation

A. Liquidity Analysis

- HDFC Bank and Dabur India consistently maintained a Current Ratio above 1, indicating solid short-term solvency. Dabur, in particular, displayed a stable and improving liquidity position over time.

- Adani Enterprises and Tata Steel showed more volatile current ratios, often below the benchmark of 1, suggesting potential strain in meeting short-term liabilities.

- NTPC, as a government PSU, demonstrated moderate but stable liquidity, often balancing between capital investment and operational expenses.
- Interpretation: Dabur and HDFC stand out for liquidity safety, while Adani and Tata Steel may need tighter working capital management.

B. Leverage (Capital Structure) Analysis

- Adani Enterprises and Tata Steel reflected high leverage levels, with elevated Debt-to-Equity and Debt-to-Assets ratios. These are capital-intensive firms and often rely on debt for expansion.
- HDFC Bank shows a unique structure with high borrowings inherent to banking operations, but it maintains stable financial gearing within regulatory norms.
- Dabur and NTPC had very low D/E ratios, showcasing conservative capital structures.
- Interpretation: While leverage is justified for asset-heavy firms like Adani and Tata Steel, Dabur's minimal reliance on debt positions it as financially prudent.

C. Profitability Ratios (Margins & Returns)

- HDFC Bank and Dabur consistently delivered high Net Profit and Return on Equity, supported by efficient operations and strong consumer demand.
- NTPC maintained steady margins but moderate profitability, reflective of its regulated utility nature.
- Tata Steel showed high margin volatility, with profits closely tied to commodity cycles.
- Adani Enterprises reported fluctuating returns, affected by its diversified business model and infrastructure-heavy investments.
- Interpretation: HDFC and Dabur are clear leaders in profitability, while NTPC offers stable but lower returns. Tata and Adani's profits are more cyclical and vulnerable to macroeconomic trends.

D. Efficiency / Activity Ratios

- Dabur and HDFC show efficient asset utilization, with consistently strong Total Asset Turnover and Inventory/Debtors Turnover.
- NTPC's and Adani's asset turnover remained low, reflecting large infrastructure investments with longer gestation periods.
- Tata Steel showed stronger inventory and asset turnover during industry up-cycles.

- Interpretation: Dabur and HDFC are leaders in operational efficiency; NTPC and Adani have long-asset cycles that slow turnover.

E. Market Valuation Ratios

- Dabur exhibited a high P/E ratio (~50x), indicating strong market confidence but also a premium valuation.
- HDFC Bank's P/E was more reasonable, aligned with its strong fundamentals.
- Tata Steel had fluctuating valuation ratios depending on the commodity cycle.
- NTPC remained undervalued by the market, despite consistent dividends, reflecting investor caution in the PSU sector.
- Adani Enterprises experienced volatile market valuations, likely influenced by investor speculation and news-driven price movements.
- Interpretation: Dabur and HDFC are investor favourites; Tata and Adani reflect cyclical and sentiment-driven valuation changes; NTPC is stable but undervalued.

F. Risk Assessment Ratios (DOL, DFL, DTL)

- Adani and Tata Steel had higher DFL/DTL, indicating increased business and financial risk due to high operating and financial leverage.
- Dabur, NTPC, and HDFC had stable and lower leverage multipliers, ensuring better risk control.
- Interpretation: Dabur and HDFC operate with low business risk; Adani and Tata carry higher operational risk due to fixed cost structures and debt obligations.

G. Overall Interpretation

- Dabur India and HDFC Bank emerge as fundamentally strong, low-risk, and high-return companies. They demonstrate consistent performance across liquidity, profitability, and efficiency metrics, backed by conservative leverage and strong market sentiment.
- NTPC is a low-risk, steady performer with government backing, suitable for stable dividend income and lower growth expectations.
- Tata Steel and Adani Enterprises show high risk–high return profiles, with profitability and valuations subject to macroeconomic and industry cycles. These companies require closer monitoring of leverage and cash flows.

3. Return and Risk Analysis of Individual Securities

3.1. Daily return calculation (1-year)

Daily returns are essential for understanding how each stock's price changes day-to-day. The formula used is the natural logarithm of the price ratio:

$$\text{Daily Return (t)} = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

- This logarithmic return is preferred because it is time-additive.
- Calculated for each trading day over one year (252 Trading Days).
- Excel Formula Used: = LN(Closing_Price_today / Closing_Price_yesterday)

This measure captures percentage changes while smoothing out compounding effects.

3.2. Annualized Risk (Standard Deviation)

The daily returns are averaged and scaled to annual numbers:

Annualized Return:

$$\text{Annualized Return} = \overline{R_{\text{daily}}} \times 252$$

Annualized Return = Average Daily Return x 252

Annualized Standard Deviation:

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$$

Annualized Std. Dev.=Std. Dev. (Daily Returns)×252Annualized.

Where 252 is the approximate number of trading days in a year. These values provide expected yearly performance and risk (volatility).

Provided in Excel as:

=AVERAGE(Annualized_Returns_Range) (For average annualized return).

=STDEV.S(Annualized_Returns_Range) (For annualized standard deviation).

3.3. Tabulated results with excel formulas

Stock	Expected Return	Daily Std. Dev.	Annualized Return	Annualized St. Dev.	Excel Formula (Daily)	Excel Formula (Annual)
NTPC	-0.1748	0.03885	=Aver.*252	=Stdev.S*SQRT(252)	=LN(P_t/P_{t-1})	=AVERAGE(E2:E251)*252, =STDEV.S(E2:E251)
Dabur India	-0.2142	0.03388	=Aver.*252	=Stdev.S*SQRT(252)	=LN(P_t/P_{t-1})	=AVERAGE(E2:E251)*252, =STDEV.S(E2:E251)
HDFC Life Insurance	0.0505	0.03692	=Aver.*252	=Stdev.S*SQRT(252)	=LN(P_t/P_{t-1})	=AVERAGE(E2:E251)*252, =STDEV.S(E2:E251)
Adani Enterprises	-0.1912	0.06706	=Aver.*252	=Stdev.S*SQRT(252)	=LN(P_t/P_{t-1})	=AVERAGE(E2:E251)*252, =STDEV.S(E2:E251)
Hindalco Industries	0.0383	0.04692	=Aver.*252	=Stdev.S*SQRT(252)	=LN(P_t/P_{t-1})	=AVERAGE(E2:E251)*252, =STDEV.S(E2:E251)

Table 2: *Tabulated Excel Results with formulas*

Interpretation: Negative expected returns indicate price declines over the period; standard deviation measures risk magnitude.

3.4. Summary insights across all 5 stocks

- NTPC and Dabur India showed negative returns, suggestive of declining prices in the year.
- HDFC Life Insurance and Hindalco had modest positive returns with relatively lower risks.
- Adani Enterprises had the highest volatility with negative return, reflecting unstable price performance.
- This implies different risk-return profiles suited to different investment strategies.

4. Portfolio Analysis: Markowitz Model

4.1. Theoretical Background

Harry Markowitz's model, introduced in 1952, is the basis for Modern Portfolio Theory (MPT). Markowitz showed that an investor can create an "efficient" portfolio by mixing asset classes to achieve the highest expected return for a given level of risk or the lowest risk for a given expected return.

Key Assumptions:

- Investors are rational and risk-averse.
- Portfolio returns are normally distributed.
- The investment decision is based only on expected return and risk (standard deviation).
- Investors prefer higher returns and lower risk.
- Assets Returns are correlated, and diversification reduces portfolio risk.

Core Concepts:

- **Expected Portfolio Return (E[Rp]):**

$$E(R_p) = \sum_{i=1}^{n} w_i E(R_i)$$

Where wi is the weight of asset iii and E(Ri) is the expected return of asset i.

- **Portfolio Variance (Risk):**

$$\sigma_p^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}$$

Where σij is the covariance between asset i and j.

Efficient Frontier:

The Markowitz Model produces a curve called the Efficient Frontier -- a set of optimal portfolios offering the highest return for the level of risk. Since portfolios below the curve are sub-optimal and portfolios above the curve are not attainable.

Benefits of Diversification:

By adding firms that possess a high bankruptcies to replace bankrupt firms in a portfolio , diversification within the financial sector can decrease the portfolio overall standard deviation more than the average of the securities' individual standard deviation, all else constant, especially if the securities are not perfectly correlated.

Limitations:

- Requires extensive computation (covariance matrix).
- Assumes a static risk-return profile and normal distribution.
- Ignores taxes, transaction costs, and liquidity issues.

4.2. Equal Weight Assumptions

In this section, the portfolio has been constructed under the Equal Weighting Assumption, which means that each of the five selected NIFTY 50 stocks are assigned an equal portfolio weight of 20% (i.e. wi = 0.20 for all i = 1 to 5).

Purpose of Equal Weighting:

- To simplify portfolio construction and make direct comparisons across assets.
- To remove bias in asset allocation and allow the analysis to focus purely on risk-return characteristics.
- To ensure uniform influence of each security on the portfolio's overall performance.

Formula Implications:

- **Portfolio Expected Return:**

$$E(R_p) = \sum_{i=1}^{n} w_i \cdot E(R_i) = \frac{1}{n} \sum E(R_i)$$

Since all wi = 0.20.

- **Portfolio Variance:**

Covariance between each pair of assets is multiplied by equal weights, simplifying the matrix calculation.

- **Portfolio Beta (Single Index Model):**

$$\beta_p = \sum w_i \cdot \beta_i = \frac{1}{n} \sum \beta_i$$

Advantages:

- Simple and intuitive.
- Promotes diversification, especially in small portfolios.
- Avoid over-reliance on forecasting models for weight optimization.

Limitations:

- Ignores market capitalization, volatility, or investor preferences.
- May overweight high-risk or underperforming assets.
- Not necessarily mean-variance optimal compared to mathematically optimized portfolios.

4.3. Calculation of portfolio expected return

The expected return of a portfolio is the weighted average of the expected returns of the individual securities in the portfolio.

Formula:

$$E(R_p) = \sum_{i=1}^{n} w_i \cdot E(R_i)$$

Where:

- E(Rp) = Expected Return of the Portfolio
- Wi = Weight of asset I in the portfolio
- E(Ri) = Expected Return of Asset i
- n = Total number of assets in the portfolio

In this report:

- The portfolio expected return of 5 NIFTY 50 stocks, each assigned equal weight of 20% (0.20).
- Expected returns for individual stocks are calculated using 1-year daily closing prices with the formula:

E (Ri) = Ln (P1/P0) * 252

- The portfolio expected return is computed as:

$$E(R_p) = \frac{1}{5} \sum_{i=1}^{5} E(R_i)$$

For a portfolio with weights wi, assigned to each stock (here equal at 20%), the expected return is the weighted sum:

$$R_p = \sum_{i=1}^{5} w_i R_i$$

= 0.2 * (-0.1748-0.2142+0.0505-0.1912+0.0383) = -0.09828 or -9.828%

4.4. Covariance & Correlation Matrix

	NIFTY50	NTPC	Dabur India	HDFC Life Insurance	Adani Enterprises	Hindalco Industries
NIFTY50	1					
NTPC	0.534675224	1				
Dabur India	0.416286878	0.753287242	1			
HDFC Life Insurance	0.75538372	0.322947212	0.0634224	1		
Adani Enterprises	0.659755577	0.88425693	0.7921458	0.427823493	1	
Hindalco Industries	0.574328517	0.62597246	0.3579441	0.485073674	0.5415577	1

Computed correlation from data analysis toolpak.

***Figure 6**: Covariance-Correlation Matrix Generated Using Correlation from Data Analysis Toolpak*

- Correlation Matrix shows relationships between stocks returns, ranging from 0 to 1.
- For example, NTPC and Adani Enterprises have a high correlation (~0.88), indicating their prices move closely.

- Covariance matrix (not shown entirely here) includes variances and covariances that quantify joint variabilities.

This is critical for diversification since assets with low or negative correlations reduce portfolio risk.

4.5. Portfolio Variance and Standard Deviation

- Portfolio Variance (Markowitz): 0.001305
- Portfolio Std. Deviation: 0.03612

Formula:

$$\text{Portfolio Variance} = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \text{Cov}(i, j)$$

(Standard deviation = SQRT(Variance))

This shows that portfolio risk is lower than the weighted average of individual risks due to diversification.

5. Portfolio Analysis: Single Index Model (SIM)

5.1. Concept and Assumptions

Concept:

The Single Index Model (SIM) from William Sharpe is a more concise way to think about portfolio analysis and asset pricing. It presumes that the return of any one stock has a linear relationship with the return of a market index (example: NIFTY 50).

The model decomposes a stock's return into:

$$R_i = \alpha_i + \beta_i R_m + e_i$$

Where:

- Ri = Return on stock i
- Rm = Return on Market Index
- αi = Stock specific return (Independent of market)
- βi = Sensitivity of stock to the market (Systematic Risk)
- ei = Unsystematic or residual error (Firm-Specific Risk)

This framework helps investors in assessing:

- Systematic risk via beta.
- Unsystematic risk via residual variance.
- Expected return using the CAPM model-like formula:

$$E(R_i) = R_f + \beta_i(R_m - R_f)$$

Key Assumptions:

- Each security's return is influenced by a single common factor – the market index.
- The relationship between the security return and the market return is linear (According to regression).
- The residual terms (errors) are uncorrelated across securities.
- The expected value of a residual term is zero: E(ei)=0
- Investors use beta as a measure of systematic risk.
- The risk-free rate and market return are known and constant over period.
- Investors are rational and seek to maximize return for a given amount of risk.

5.2. Regression Output: Alpha, Beta, R^2, Residual risk

To implement the Single Index Model, we perform a linear regression of the stock's daily returns Ri against the markets daily returns Rm. (NIFTY 50 in our case).

$$R_i = \alpha + \beta R_m + e$$

Using excel, the output of this regression gives key components of the SIM:

- **Beta (β) – Systematic Risk:**

Definition: Measures the sensitivity of the individual stock's return to the market movements.

Interpretation:

  - $\beta = 1$: Moves in line with the market.
  - $\beta > 1$: More volatile than the market.
  - $\beta < 1$: Less volatile than the market.

From Regression: The slope coefficient of the egression line.

- **Alpha (α) – Stock Specific Return:**

Definition: The intercept of the regression line which measures the excess return which is not explained by the market movement.

Interpretation:

  - $\alpha > 0$: Means the stock outperforms the market adjusted returns.
  - $\alpha < 0$: Means the stock underperforms the market adjusted returns.

From Regression: The value of intercept.

- **R-Squared (R^2) – Goodness of Fit**

Definition: The proportion of the stocks return variability which is explained by the market.

Interpretation:

  - $R^2 = 0.80$: That means 80% of the stocks movements is explained by the market.
  - Higher the R^2 indicates a stronger market linkage, whereas a lower R^2 indicates more firm-specific factors.

From Regression: Shown as R square when the computation of regression is done using the excel data analysis tool-pack.

- **Residual Risk (Unsystematic Risk) – Firm-specific Risk**

Definition: The risk that is not explained by market movements, also referred to as idiosyncratic risk.

Formula:

  - Residual Mean Square (MSE) = SS Residual / df = σ_e^2
  - Residual Risk (Standard Deviation) = $\sqrt{\sigma_e^2}$

From Regression: In excels ANOVA table, use:

  - Residual SS = Residual Sum of Squares
  - DF = Residual Degrees of Freedom

5.3. Expected Return (Formula Based)

Expected Return of a Stock:

The expected return measures the mean anticipated return of an asset, based on past performance.

$$E(R_i) = \frac{1}{N}\sum_{t=1}^{N} r_{i,t}$$

Where:

- E(Ri): Expected return of asset i*i*
- N: Number of periods (e.g., trading days)
- ri*t*: Daily return on day t

Annualized Expected Return:

$$E(R_{i,annual}) = \text{Average Daily Return} \times 252$$

Excel Formula:

- Daily expected: =AVERAGE(Returns_Range)
- Annualized: =AVERAGE(Returns_Range)*252

Security Name	Expected Returns (Individual)
NTPC	-17.4816019%
Dabur India	-21.4247254%
HDFC Life Insurance	5.05206051%
Adani Enterprises	-19.1235966%
Hindalco Industries	3.82938669%

Table 3: Expected Returns of Individual Stocks

5.4. Portfolio Return, Beta, Risk

Portfolio Return

For a portfolio of n stocks:

$$E(R_p) = \sum_{i=1}^{n} w_i \cdot E(R_i)$$

Where:

- E(Rp): Expected portfolio return
- wi: Portfolio weight of stock i

Equal Weights (0.2 each)

E(*Rp*) = 0.2 * (-0.1748-0.2142+0.0505-0.1912+0.0383) = -0.09828 or -9.828%

Excel Formula: =SUMPRODUCT(Expected_Returns_Range, Weights_Range)

Portfolio Beta:

Portfolio beta is the weighted sum of the individual betas:

$$\beta_p = \sum_{i=1}^{n} w_i \cdot \beta_i$$

Betas (from regression sheet):

- NTPC: 1.112
- Dabur India: 0.603
- HDFC Life: 0.802
- Adani Ent.: 1.578
- Hindalco: 1.313
- **Portfolio Beta: 1.081**

Excel Formula: =slope(Array1_Array2)

Portfolio Risk:

Markowitz (Variance-Covariance):

$$\sigma_p^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \mathrm{Cov}(i, j)$$

- Calculated using both individual variances and covariances among assets.
- **Portfolio Variance (from sheet):** 0.00130
- **Portfolio Std. Deviation:** SQRT(0.00130) ≈ 0.0361

Single Index (Systematic + Diversifiable risk):

$$\sigma_p^2 = \beta_p^2 \cdot \sigma_m^2 + \sum_{i=1}^{n} w_i^2 \cdot \sigma_{\epsilon,i}^2$$

Values (from sheet):

- Portfolio Risk: 4.90×10−54.90×10−5
- Portfolio Std. Deviation: 0.00700.0070

Excel Formula:

- For Markowitz: Use variance-covariance matrix and =MMULT/=SUMPRODUCT
- For Single Index: Combine squared beta, market variance, and residual risks

5.5. Interpretation of results

Expected Return:

- **HDFC Life Insurance** and **Hindalco Industries** provided modest positive annualized expected returns. Other stocks delivered negative returns, indicating underperformance over the year.

Portfolio Metrics:

- **Portfolio Return** is *negative* (-9.8%), as underperformers outweigh positive contributors.
- **Portfolio Beta** is *slightly above 1* (1.08), so the portfolio is nearly as sensitive as the broader market.
- **Portfolio Risk** (Standard Deviation) is *lower than most individual stock risks* due to diversification—showing risk reduction even though returns are weak.

Takeaway:

- The portfolio is diversified (mixing lower and higher betas, and different sectors), which dampens volatility.
- However, asset selection is critical—picking several underperformers can yield negative returns even in a risk-controlled portfolio.

6. Model Comparison and Interpretation

6.1. Portfolio metrics from both models

Key Output metrics for our 5-stock portfolio using Markowitz Model and Single Index Model Comprises of the following:

Metric	Markowitz Model	Single Index Model (SIM)
Expected Portfolio Return	-9.82970%	-0.0295469%
Portfolio Standard Deviation (Risk)	3.612215%	0.700092942%
Portfolio Beta	Not Used	1.08142622
Portfolio Alpha	Not Used	-0.000389835
Covariance Matrix Used	Yes	No (Uses Market Return only)
Time Period	1 year (Daily Data)	1 year (Daily Data)
Weighting Method	Equal Weight (20% each)	Equal Weight (20% each)

Table 4: Output Metrics Comparison between Markowitz and Single Index Model

- Markowitz Model: uses full covariance matrix for mean-variance optimization.
- Single Index Model: uses regression of stocks against NIFTY50 index, characterized by Alpha, Beta, Residual variance.
- Portfolio Beta (Single Index) ≈ 1.081, indicating slightly higher sensitivity than market.
- Portfolio Alpha (Single Index) ≈ -0.00039, indicating slight underperformance vs market risk.
- Portfolio risk (Single Index std. dev.) ≈ 0.0070 (daily or smaller scale) showing lower risk estimate compared to Markowitz

6.2. Strengths/limitations of Markowitz vs SIM

A. Markowitz Model

- **Strengths:**
  - Considers all pairwise covariances, offering the most theoretically sound risk-return trade-off.
  - Enables identification of the efficient frontier of optimal portfolios.
- **Limitations:**
  - Computationally heavy: requires **n(n-1)/2 covariances** for *n* assets.
  - **Sensitive to input errors** in expected returns and covariances.
  - Not scalable for large portfolios without optimization tools.

B. Single Index Model

- **Strengths:**
  - **Simplified Model:** Only need a single index (Eg: NIFTY 50) to estimate covariances.

- Easier and faster to implement using regression analysis.
- Good for practical use when time or data are limited.

- **Limitations:**
  - Assumes market return is the only factor affecting asset returns (which is unrealistic).
  - May underestimate risk when stocks are affected by other factors (sector specific shocks, geopolitical risks).
  - Alpha and residual errors can distort predictions if not stable.

6.3. Risk-return trade-off insight

- **Markowitz Model:** Captures all pairwise relationships, providing more precise risk estimation and better optimization.
- **Single Index Model:** Simplifies computation but may underestimate risk due to ignoring specific covariances.
- Here, both models indicate negative expected returns with risk somewhat controlled by diversification.

6.4. Which model better fits your portfolio

- Markowitz is superior for portfolios with multiple assets and visible correlations; achieves better diversification.
- Single Index is simpler and useful for quick estimates, especially when market factor dominates.
- Because correlations are moderate to high here, Markowitz provides a more accurate and actionable picture.

7. Key Observations and Investment Insights

7.1. Best-performing stock

- **HDFC Life Insurance:** Positive expected return (0.0505), with a moderate risk (0.0369), outperforming in risk-adjusted terms.
- **Hindalco Industries:** Moderately positive return (0.0383), slightly higher risk.
- Others like NTPC, Dabur, and Adani gave negative returns, making them less attractive from performance perspective.

7.2. Diversification benefits

Diversification reduces total portfolio risk—portfolio volatility (0.0361–0.0070) is less than the average individual standard deviations (ranging 0.0339–0.0671).

Negative or low correlations help absorb adverse shocks, as seen in the reduced overall variance and standard deviation compared to individual stocks.

7.3. Investor takeaway: Conservative vs Aggressive Portfolios

- Conservative: Choose assets with positive returns and lower risks (e.g., HDFC Life, Hindalco).
- Aggressive: May include higher-volatility stocks like Adani Enterprises hoping for turnarounds, but that's not rewarded here.
- Markowitz allows optimizing this mix to match risk preferences; Single Index helps understand market risk exposures.

7.4. Stock-specific recommendations

- HDFC Life Insurance and Hindalco Industries: Prefer for stable, modest growth in a conservative portfolio.
- NTPC and Dabur India: Avoid at present given negative expected returns.
- Adani Enterprises: High risk, negative mean—only for risk-tolerant investors with strong conviction in turnaround.

Summary:

- Daily returns and annualized metrics provide the foundation for portfolio evaluation.
- Markowitz model's covariance-aware approach gives better insight and diversification benefits.
- Single Index model simplifies market-related systematic risk but lacks precision in portfolio variance.
- Current portfolio suggests losses overall, highlighting the importance of vigilant stock selection.
- Diversification notably reduces portfolio risk.
- Investors should align portfolio choice with risk capacity and market outlook, preferring stable performers or selectively adding volatile stocks per their strategy.

8. References

1] *Adani Enterprises profit rockets on growing airports and green business*. (n.d.). Retrieved July 27, 2025, from https://www.ft.com/content/c9fe5f16-ee48-46f0-98ea-3c13cc09765f

2] *Businesses | Adani Enterprises Ltd*. (n.d.). Retrieved July 27, 2025, from https://www.adanienterprises.com/businesses

3] *Dabur India Ltd Company Profile - Dabur India Ltd Overview - GlobalData*. (n.d.). Retrieved July 29, 2025, from https://www.globaldata.com/company-profile/dabur-india-ltd/

4] HDFC Life – One of the Best Life Insurance Companies in India. (n.d.). *Www.Hdfclife.Com*. Retrieved July 27, 2025, from https://www.hdfclife.com/about-us/hdfc-life-introduction

5] *Hindalco Industries Ltd: Leading Aluminium & Copper Producer*. (n.d.). Retrieved July 27, 2025, from https://www.adityabirla.com/businesses/companies/hindalco-industries-limited

6] **Hindalco's ₹45,000 Crore Investment Plan* : r/FuturesFundamentals*. (n.d.). Retrieved July 27, 2025, from https://www.reddit.com/r/FuturesFundamentals/comments/1jgd9b7/hindalcos_45000_crore_investment_plan

7] *NTPC Overview | NTPC Limited*. (n.d.). Retrieved July 27, 2025, from https://ntpc.co.in/ntpc-overview?utm_source

8] *World's Best Ayurvedic Brand: Plastic Waste Positive FMCG Company | Dabur*. (n.d.). Retrieved July 29, 2025, from https://www.dabur.com/